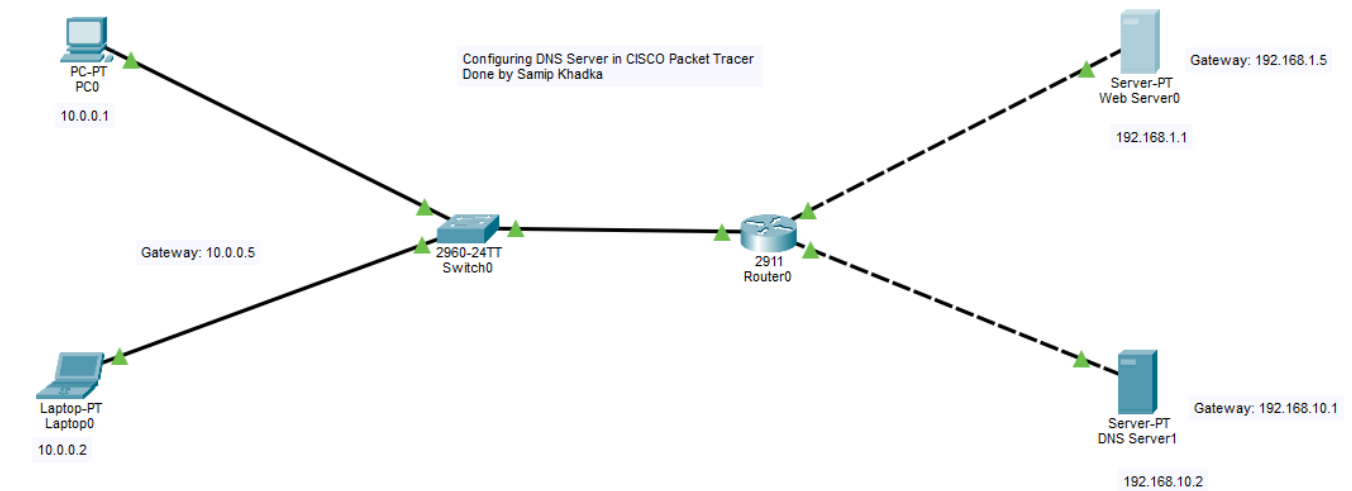




PC0

Physical	Config	Desktop	Programming	Attributes
IP Configuration				
Interface		FastEthernet0		
IP Configuration				
☐ DHCP		☒ Static		
IPv4 Address		10.0.0.1		
Subnet Mask		255.0.0.0		
Default Gateway		10.0.0.5		
DNS Server		192.168.10.2		

Laptop0

Physical	Config	Desktop	Programming	Attributes
IP Configuration				
Interface		FastEthernet0		
IP Configuration				
☐ DHCP		☒ Static		
IPv4 Address		10.0.0.2		
Subnet Mask		255.0.0.0		
Default Gateway		10.0.0.5		
DNS Server		192.168.10.2		

Web Server0

Physical	Config	Services	Desktop	Programming
IP Configuration				
IP Configuration				
☐ DHCP		☒ Static		
IPv4 Address		192.168.1.1		
Subnet Mask		255.255.255.0		
Default Gateway		192.168.1.5		
DNS Server		192.168.2.5		

DNS Server1

Physical	Config	Services	Desktop	Programming
IP Configuration				
IP Configuration				
☐ DHCP		☒ Static		
IPv4 Address		192.168.10.2		
Subnet Mask		255.255.255.0		
Default Gateway		192.168.10.1		
DNS Server		0.0.0.0		

Router0

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

GigabitEthernet0/2

GigabitEthernet0/0

Port Status

☒ On

Link Speed

☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex

☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address

00E0.8FDE.0D01

IP Configuration

IPv4 Address

10.0.0.5

Subnet Mask

255.0.0.0

Tx Ring Limit

10

Router0

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

GigabitEthernet0/2

GigabitEthernet0/1

Port Status

☒ On

Link Speed

☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex

☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address

00E0.8FDE.0D02

IP Configuration

IPv4 Address

192.168.1.5

Subnet Mask

255.255.255.0

Tx Ring Limit

10

Router0

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

GigabitEthernet0/2

GigabitEthernet0/2

Port Status

☒ On

Link Speed

☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex

☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address

00E0.8FDE.0D03

IP Configuration

IPv4 Address

192.168.10.1

Subnet Mask

255.255.255.0

Tx Ring Limit

10

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS**
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

DNS

DNS Service ☒ On ☐ Off

Resource Records

Name Type **A Record**

Address

Add Save Remove

No.	Name	Type	Detail
0	samip.com	A Record	192.168.1.1

Physical Config **Services** Desktop Programming Attributes

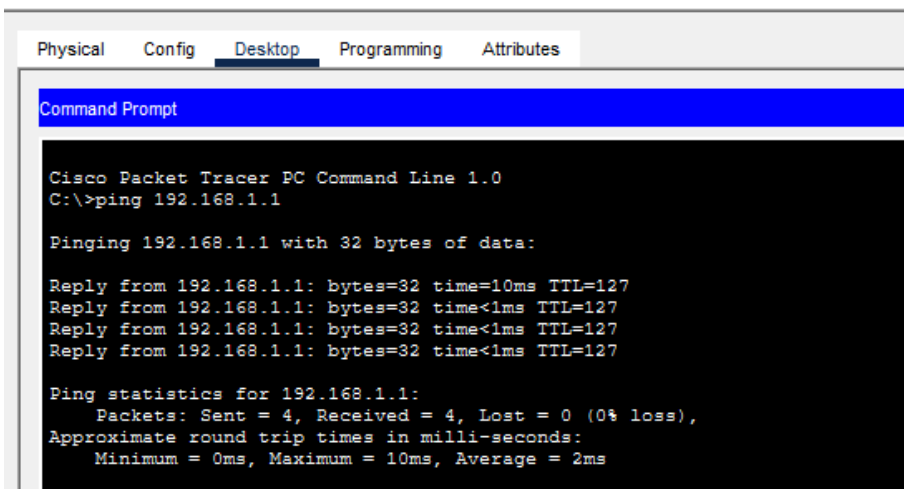
SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

File Name:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>All About DNS</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"
rel="stylesheet">
  <style>
    /* Custom CSS to complement Bootstrap */
    body {
      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
      background-color: #f8f9fa;
      color: #333;
    }
    .hero-section {
      background: linear-gradient(135deg, #0d6efd, #003b95);
      color: white;
      padding: 60px 0;
      border-bottom: 5px solid #002359;
    }
    .card {
      border: none;
      box-shadow: 0 4px 6px rgba(0,0,0,0.1);
      transition: transform 0.2s;
    }
    .card:hover {
      transform: translateY(-5px);
    }
    footer {
      background-color: #212529;
      color: #ced4da;
      padding: 30px 0;
      margin-top: 60px;
    }
  </style>
</head>
<body>

  <header class="hero-section text-center">
    <div class="container">
      <h1 class="display-3 fw-bold">Domain Name System (DNS)</h1>
      <p class="lead">The Phonebook of the Internet</p>
    </div>
  </header>
```



Physical Config Desktop Programming Attributes

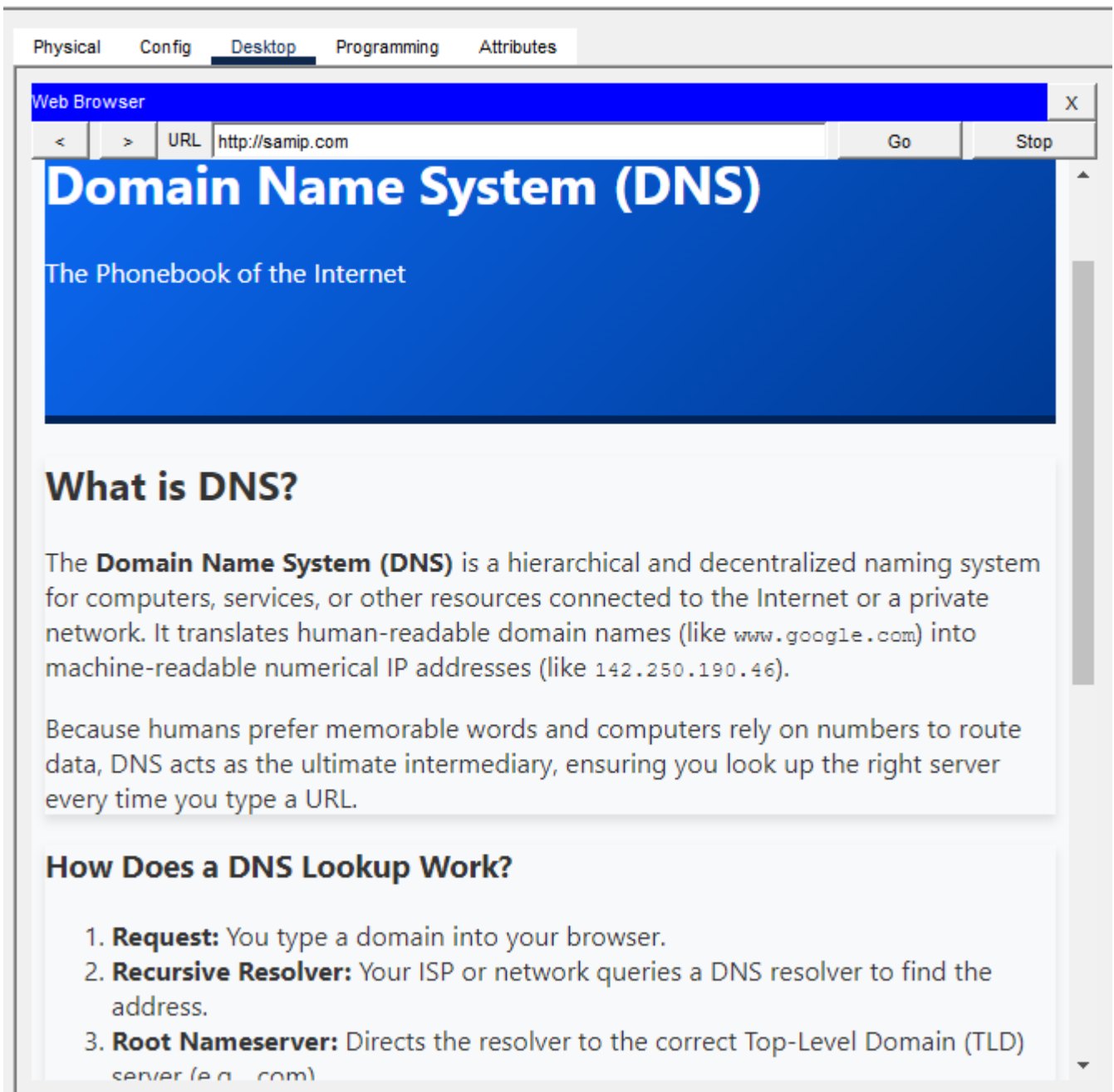
Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=10ms TTL=127
Reply from 192.168.1.1: bytes=32 time<1ms TTL=127
Reply from 192.168.1.1: bytes=32 time<1ms TTL=127
Reply from 192.168.1.1: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms
```



Physical Config Desktop Programming Attributes

Web Browser

< > URL <http://samip.com> Go Stop

Domain Name System (DNS)

The Phonebook of the Internet

What is DNS?

The **Domain Name System (DNS)** is a hierarchical and decentralized naming system for computers, services, or other resources connected to the Internet or a private network. It translates human-readable domain names (like www.google.com) into machine-readable numerical IP addresses (like 142.250.190.46).

Because humans prefer memorable words and computers rely on numbers to route data, DNS acts as the ultimate intermediary, ensuring you look up the right server every time you type a URL.

How Does a DNS Lookup Work?

1. **Request:** You type a domain into your browser.
2. **Recursive Resolver:** Your ISP or network queries a DNS resolver to find the address.
3. **Root Nameserver:** Directs the resolver to the correct Top-Level Domain (TLD) server (e.g., com).